Ⅲ. Direct limit (without the language of category theory)

Def Ⅲ.1.1. Let I be a directed set. A a ring. $(M_i)_{i\in I}$ a family of A-modules.

$\mu_{ij}: M_i \to M_j$ homomorphism. satisfying

(1) $\mu_{ii} = id_{M_i}$; (2) $\mu_{ik} = \mu_{jk} \circ \mu_{ij}$ $\forall i \le j \le k$.

$\mathbb{M} = (M_i, \mu_{ij})$ is called a direct system. over I. $l_i: M_i \to \bigoplus_{i\in I} M_i$ embedding.

Let $C = \bigoplus_{i\in I} M_i$. D be the submodule of C generated by elements: $l_i(x_i) - l_j\mu_{ij}(x_i)$ $(i \le j)$ $x_i \in M_i$

let $M = C/D$. $\mu: C \to M$ be the projection and let $\mu_i = \mu|_{M_i}: M_i \to M$

Then (M, μ_i) is called the direct limit of $\mathbb{M}$., written $\varinjlim M_i$.

Prop. Ⅲ.1.2. (i). $\forall x \in \varinjlim M_i$, $\exists$ $i \in I$. $x_i \in M_i$. s.t. $x = \mu_i(x_i)$.

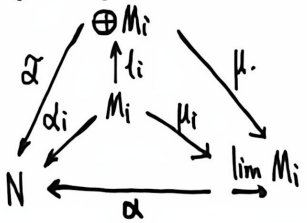
(ii) If $\mu_i(x_i) = 0$. then $\exists j \ge i$. s.t. $\mu_{ij}(x_i) = 0$.

pf: (skipped) $\cdots \to M_i \xrightarrow{\mu_{ij}} M_j \xrightarrow{\mu_{jk}} M_k \to \cdots$

Prop. Ⅲ.1.3. (the UMP of $\varinjlim M_i$). N: R-module $\alpha_i: M_i \to N$ homomorphism.

$\forall i \le j$. $\alpha_i = \alpha_j \circ \mu_{ij}$ then $\exists!$ $\alpha: \varinjlim M_i \to N$, $\alpha_i = \alpha \circ \mu_i$ $\forall i \in I$

pf: Consider the diagram:



Some specific examples:

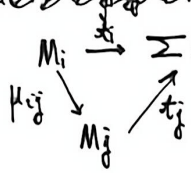
① $(M_i)_{i\in I}$ be a family of submodules of an A-module, s.t. $\forall i,j\in I\ \exists k\in I$.

s.t. $M_i+M_j\subseteq M_k$. $i\le j \Leftrightarrow M_i\subseteq M_j$ $\mu_{ij}: M_i\to M_j$ be the embedding.

then $\varinjlim M_i = \sum M_i = \bigcup M_i$.

In particular, any A-module is the direct limit of its finitely generated submodules.

pf:



t_i embedding $\forall i\le j$ we have $t_i = t_j\cdot\mu_{ij}$.

Then by the UMP of the direct limit.

We have $\alpha: \varinjlim M_i \to \sum M_i$ isomorphism

$\Rightarrow \varinjlim M_i = \sum M_i = \bigcup M_i$ (the last one is similar) ▨

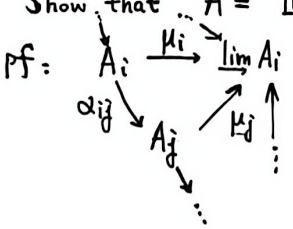
② Let $(A_i)_{i\in I}$ be a family of rings. $\alpha_{ij}: A_i\to A_j$ a ring homomorphism.

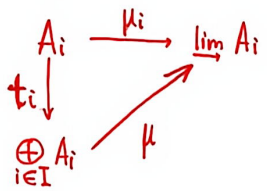
satisfying (i) $\alpha_{ii} = id_{A_i}$,

(ii) $\alpha_{ik} = \alpha_{jk}\cdot\alpha_{ij}$ $(i\le j\le k)$.

Show that $A = \varinjlim A_i$ is a ring. $A_i\to A$. are ring homomorphism.

pf:





$\varinjlim A_i$ is a $\mathbb{Z}$-module. $= \mu_k(\mu_{ik}(x_i))$ $\mu_k(\mu_{jk}(x_j))$

$\forall (x,y)\in \varinjlim A_i\times\varinjlim A_i$ $x=\mu_i(x_i)$, $y=\mu_j(x_j)$

Define multiplication: $\varinjlim A_i\times\varinjlim A_i \to \varinjlim A_i$

$(x,y)\mapsto \mu_k(\mu_{ik}(x_i)\cdot\mu_{jk}(x_j))$

~~In particular take $(A_i)_{i\in\mathbb{N}}=(\mathbb{Z})_{i\in\mathbb{N}}$, $\alpha_{ij}:\mathbb{Z}\to\mathbb{Z}$~~

~~$x'\cdot y'$~~ the 1 in $\varinjlim A_i := \mu_k(1_k)$ 1_k is identity in A_k.

~~THEN Thm~~

By caculation, we have A is a ring and $\mu_i: A_i\to A$ ring homomorphism.

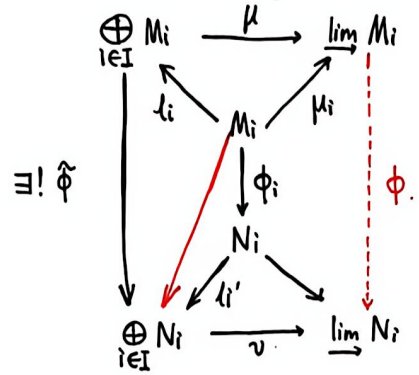
allay III. 1.3. $nil(A_i)$ be the nilpotent ~~elements~~ set. in A_i. Then $\varinjlim nil(A_i)$ ~~are~~ is nilpotent set in $\varinjlim A_i$

$\mathbb{M} = (M_i, \mu_{ij})$, $\mathbb{N} = (N_i, \nu_{ij})$ be direct systems of A-modules over I

A homomorphism $\Phi: \mathbb{M} \to \mathbb{N}$ is a family of A-module homomorphisms $\phi_i: M_i \to N_i$ s.t. $\phi_j \circ \mu_{ij} = \nu_{ij} \circ \phi_i \quad \forall i \le j$.

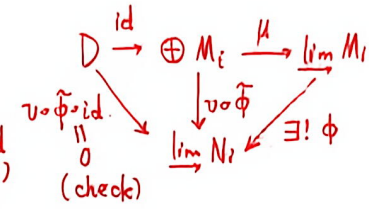
Then we have a unique homomorphism $\phi = \varinjlim \phi_i : \mathbb{M} \to \mathbb{N}$ s.t. $\phi \circ \mu_i = \nu_i \circ \phi_i \quad \forall i \in I$

pf: Consider the following commute diagram: and note that



$\mathrm{Coker}(\mu) = \varinjlim M_i$

$\mathrm{Coker}(\nu) = \varinjlim N_i$



Then by the UMP of $\mathrm{Coker}(\mu)$ (id) we just need to ~~identify~~ check

$\nu \circ \tilde{\Phi} \left(l_i(x_i) - l_j \circ \mu_{ij}(x_i) \right) = 0$ ▨

Prop. III.1.5. A sequence of direct systems and homomorphisms $\mathbb{M} \to \mathbb{N} \to \mathbb{P}$ is exact if the corresponding sequence of modules and module homomorphisms is exact $\forall i \in I$.

Show that: $\mathbb{M} \to \mathbb{N} \to \mathbb{P}$ of direct limits is then exact.

pf: (skipped). ▨

Prop. III.1.6. R ring. $(M_i)_{i\in I}$ R-modules. R-module N. Then $(\oplus M_i) \otimes N \cong \oplus (M_i \otimes N)$

Prop. III.1.7. $f: N' \to N$ R-module homomorphism. then $\forall$ R-module M, $M \otimes \mathrm{Coker}(f) \cong \mathrm{Coker}(1_M \otimes f)$.

Thm III.1.8. $\mathbb{M} = (M_i, \mu_{ij})$. $\mu_i: M_i \to M$; N: R-module. Then $(M_i \otimes N, \mu_{ij} \otimes 1)$ is a direct system, and $(\varinjlim M_i) \otimes N \cong \varinjlim (M_i \otimes N)$.

pf: let $l_i: M_i \to \oplus M_i$ $\quad l_i': M_i \otimes N \to \oplus (M_i \otimes N)$

then

$$\begin{array}{ccccccc} \mathrm{Ker}(\mu') & \xrightarrow{l'} & \oplus (M_i \otimes N) & \xrightarrow{\mu'} & \varinjlim (M_i \otimes N) & \to & 0 \\ & & \varphi \uparrow \downarrow \psi & & & & \\ \mathrm{Ker}(\mu \otimes 1_N) & \xrightarrow{l \otimes 1_N} & (\oplus M_i) \otimes N & \xrightarrow{\mu \otimes 1_N} & (\varinjlim M_i) \otimes N & \to & 0 \end{array}$$

▨

Corollary III.1.9. M_i $(i \in I)$ are R-modules. $M := \oplus M_i$ then M flat $\iff \forall i \in I$. M_i flat.

pf: Consider short exact sequence ▨

Corollary III.1.10. A-module M, $\mathfrak{a}$ is an ideal. Then $(A/\mathfrak{a}) \otimes_A M \cong M/\mathfrak{a}M$

pf: (skipped) ▨

2 Presentation of modules and Projective modules

Def III 2.1 The presentation of M is the exact sequence:

$G \to F \to M \to 0$ (G, F free)

If $\mathrm{rank}(G), \mathrm{rank}(F) < \infty$ Then call it's finite.

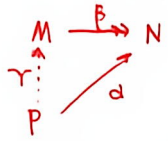
M finitely presentation

Prop. III 2.2. R ring, M is a R-module. $m_\lambda (\lambda \in \Lambda)$ be the generator. Then we have $0 \to K \to R^{\oplus \Lambda} \xrightarrow{\alpha} M \to 0$

where $\alpha(e_\lambda) = m_\lambda$. $\{e_\lambda\}$ be the standard basis, and $R^{\oplus \Sigma} \to R^{\oplus \Lambda} \to M \to 0$.

Def III. 2.3. P is a projective module. if $\forall$ surjective $\beta: M \to N$. every $\alpha: P \to N$ can be lifted as:

$\gamma: P \to M$. i.e. $\alpha = \beta\gamma$.



Prop. III.2.4. $R^{\oplus \Lambda}$ are projective modules.

Def. III.2.5. (i). $\rho: M \to M'$ is the retraction of $\alpha: M' \to M$, if $\rho\alpha = 1_{M'}$

(ii). $\sigma: M'' \to M$ is the section of $\beta: M \to M''$, if $\beta\sigma = 1_{M''}$

(iii). exact sequence: $M' \xrightarrow{\alpha} M \xrightarrow{\beta} M''$ is split if $\exists\, \varphi$ isomorphism: $M \to M' \oplus M''$

s.t. $\varphi\alpha = \iota_{M'}$ $\beta = \pi_{M''} \circ \varphi$.

Prop. III.2.6. $M' \xrightarrow{\alpha} M \xrightarrow{\beta} M''$ exact. TFAE:

(i) the exact sequence split;

(ii) $\exists$ the retraction of α: $\rho: M \to M'$. and β is surjective.

(iii) $\exists$ the section of β: $\sigma: M'' \to M$ and α is injective.

pf: (skipped).

Thm III.2.7. R-module P, TFAE:

(i) P is projective

(ii). $\forall$ short exact sequence $0 \to K \to M \to P \to 0$ split

(iii) $\exists$ R-module K. s.t. $K \oplus P$ free module.

(iv) $\forall$ exact sequence: $N' \to N \to N''$ we have $\mathrm{Hom}(P, N') \to \mathrm{Hom}(P, N) \to \mathrm{Hom}(P, N'')$ exact.

(v) $\forall\, \beta: M \twoheadrightarrow N$. introduces: $\mathrm{Hom}(P, \beta): \mathrm{Hom}(P, M) \to \mathrm{Hom}(P, N)$ surjective.

(Schanuel) P, P' are projective. $\forall$ exact sequences are isomorphism

$$\begin{array}{ccccccccc} 0 & \to & L & \xrightarrow{i} & P & \xrightarrow{\alpha} & M & \to & 0 \\ & & & & \downarrow\beta & & & & \\ 0 & \to & L' & \xrightarrow{i'} & P' & \xrightarrow{\alpha'} & M & \to & 0 \end{array} \Rightarrow \begin{array}{ccccccccc} 0 & \to & L\oplus P' & \xrightarrow{i\oplus 1_{P'}} & P\oplus P' & \xrightarrow{(\alpha,0)} & M & \to & 0 \\ & & \downarrow\simeq\rho & & \downarrow\simeq\gamma & & \downarrow=1_M & & \\ 0 & \to & P\oplus L' & \xrightarrow{1_P\oplus i'} & P\oplus P' & \xrightarrow{(0,\alpha')} & M & \to & 0 \end{array}$$

Pf: Construct

$$\begin{array}{ccccccccc} 0 & \to & L\oplus P' & \xrightarrow{i\oplus 1_{P'}} & P\oplus P' & \xrightarrow{(\alpha\ 0)} & M & \to & 0 \\ & & \simeq\uparrow\lambda & & \uparrow\theta & & \uparrow=1_M & & \\ 0 & \to & K & \longrightarrow & P\oplus P' & \xrightarrow{(\alpha\ \alpha')} & M & \to & 0 \\ & & \| & & & & & & \\ & & \ker(\alpha\ \alpha') & & & & & & \end{array}$$

where $\theta = \begin{pmatrix} 1 & \pi \\ 0 & 1 \end{pmatrix}$ α surjective. $\exists\, \pi: P' \to P$ s.t. $\alpha' = \pi\alpha$ ▨

Lemma III.2.9. R ring, exact sequence: $0 \to L \to R^n \to M \to 0$ Then M finitely represented $\Leftrightarrow$ L finitely generated.

Lemma III.2.10. R ring, $0 \to L \xrightarrow{\alpha} M \xrightarrow{\beta} N \to 0$ exact. L finitely generated, M finitely represented then $\Leftrightarrow$ N finitely represented.

Prop. III.2.11 R ring, $0 \to L \xrightarrow{\alpha} M \xrightarrow{\beta} N \to 0$ exact, L, N finitely represented. then M finitely ~~generate~~ represented.

3. Caley-Hamilton thm.

Thm. III.3.1. R-module M generated by $m_1, \dots, m_n$. $\varphi: M \to M$. $m_i \mapsto \sum_{j=1}^{n} a_{ij} m_j$ $a_{ij} \in R$.
$M = (a_{ij})$. let P_M be its characteristic polymial.
Then $P_M(\varphi) = 0$

pf: let $\mu_{a_{ij}}: R \to R$. $x \mapsto a_{ij}x$. $\Delta := (\delta_{ij}\varphi - \mu_{a_{ij}})$. $X := (m_1, \dots, m_n)^T$

Then $\Delta X = 0$ $\Rightarrow 0 = \Delta^* \Delta X = \det(\Delta) X$ $\Rightarrow \det(\Delta) m_j = 0$ $1 \le j \le n$ $\Rightarrow P_M(\varphi) = 0$ ▨

Prop. III.3.2. M finitely generated R-module. a is an ideal. Then $M = aM \Leftrightarrow \exists\, x \in a$. s.t. $(1+x)M = 0$

pf: $\Rightarrow$: Let $m_1, \dots, m_n$ be the generator, $m_i = \sum_{j=1}^{n} a_{ij} m_j$ $a_{ij} \in a$. $M = (a_{ij})$.
$P_M(T) = T^n + a_1 T^{n-1} + \dots + a_n$ let $x = (a_1 + \dots + a_n) \in a$. ϕ be the identity.
$\Rightarrow (1+x)M = 0$

Cora III.3.3. M fr-R-module. $\varphi \in \mathrm{End}(M)$. If φ surjective. then φ isomorphism.

Lemma III.3.4. (Nakayama) R M fr-R-module, $m \subseteq \mathrm{rad}(M)$. and $mM = M$. Then $M = 0$.

Prop. III.3.5. $N \subseteq M$ submodule, $m \subseteq \mathrm{rad}(M)$ ideal.

(1) If M/N fr-R-module. and $N + mM = M$. then $N = M$.

(2). If M fr-R-module. then $m_1, \dots, m_r$ generate M $\Leftrightarrow$ $\pi: M \to M/mM$, $\pi(m_i)$ generate M/mM